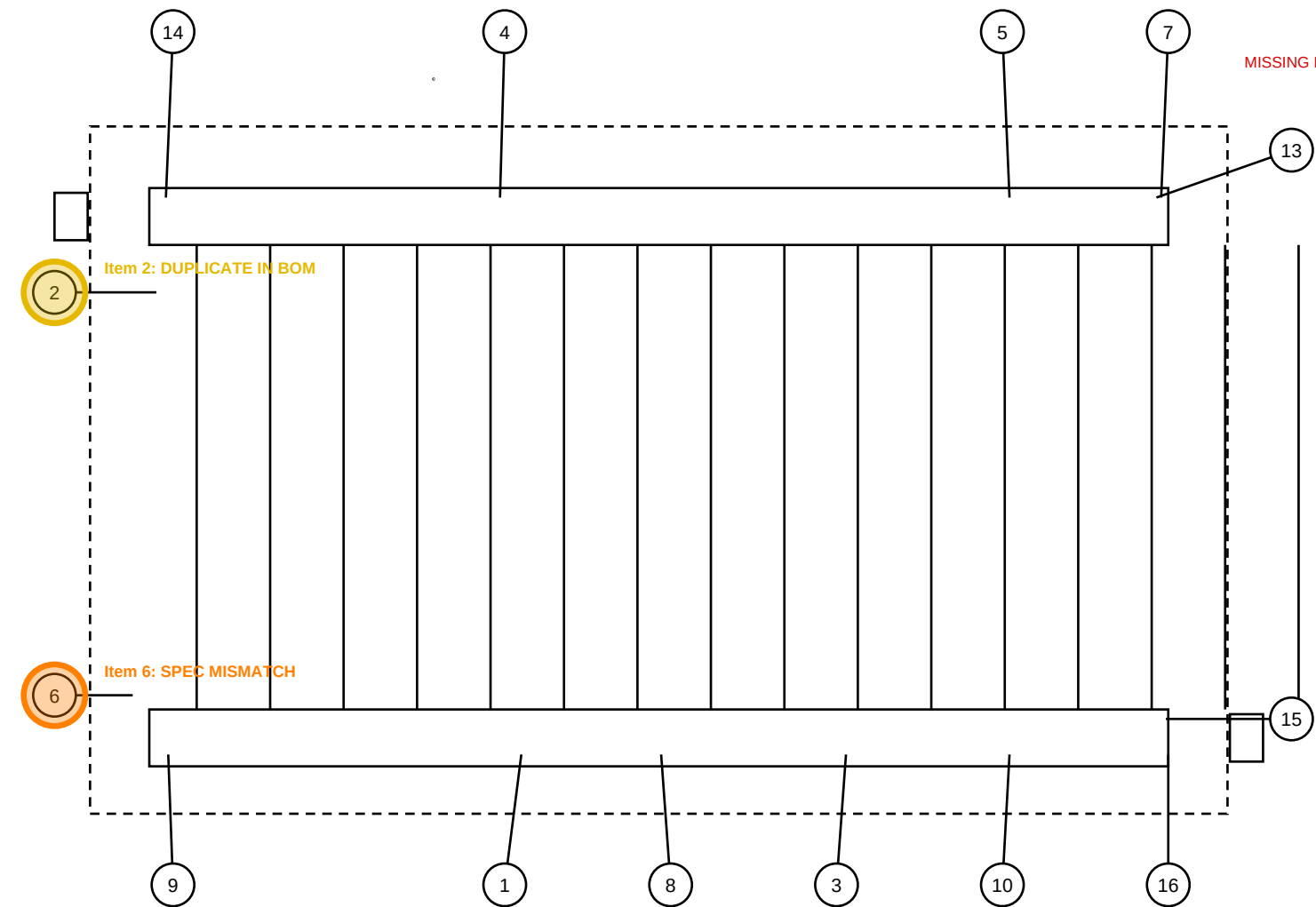


BOM CHECK (automated - verify before acting)

- red = on drawing, missing in BOM
- orange = qty / spec / revision mismatch
- yellow = confirm match / extra / duplicate
- blue = suggestion (implied item, rule-based)



Item	Description	Material	Qty	Part No	Remarks
1	INLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1	EC-01-01	
2	OUTLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1	EC-01-02	
3	ECONOMISER TUBE 38.1 OD X 3.6 THK	SA 210 GR.A1	140	EC-01-03	
4	RETURN BEND 38.1 OD	SA 210 GR.A1	24	EC-01-04	
5	TUBE SUPPORT PLATE 12 THK	IS 2062 GR.B	11	EC-01-05	
6	INLET FLANGE 200 NB CL.150	SA 105	1	EC-01-06	
7	OUTLET FLANGE 200 NB CL.150	SA 105	1	EC-01-07	
8	DRAIN VALVE 25 NB	SA 105	2	EC-01-08	
9	VENT VALVE 15 NB	SA 105	1	EC-01-09	
10	THERMOWELL 1/2 NPT	SS 316	2	EC-01-10	
11	GASKET 200 NB CL.150	SS 304 + GRAPHITE	2	EC-01-11	
12	STUD BOLT M20 X 90 WITH 2 NUTS	SA 193 B7	16	EC-01-12	
13	LIFTING LUG 16 THK	IS 2062 GR.B	4	EC-01-13	
14	CASING PLATE 3 THK	IS 2062	18	EC-01-14	
15	EXPANSION JOINT 600 X 600	FABRIC	1	EC-01-15	
16	INSPECTION DOOR 300 X 300	IS 2062	2	EC-01-16	

NOTES:

- ALL DIMENSIONS ARE IN MM.
- ALL WELDING AS PER IBR / ASME SEC I.
- INSULATE WITH 100 THK MINERAL WOOL AND CLAD WITH 0.6 THK ALUMINIUM SHEET.

BOM LACKS: Cladding

ECONOMISER ASSEMBLY

DRG NO : EC-01-GA-001
DRG REV : B
SCALE : NTS SIZE : A3
SAMPLE (SYNTHETIC) DRAWING

BOM CHECK SUMMARY

Drawing : sample_01_drawing.pdf (Rev B)
BOM : sample_01_bom.xlsx (Rev B)
Input : vector PDF
RUN RELIABILITY : HIGH

Drawing items checked : 16
BOM lines read : 18
BOM completeness : 93.8% (drawing items that have a BOM line)

Findings: duplicate 1, implied 1, omission 1, spec 1

[RED] Item 12 'STUD BOLT M20 X 90 WITH 2 NUTS' (qty 16) is on the drawing but has no BOM line
[ORANGE] Item 6: specification mismatch - dimension: drawing ['200'] vs BOM ['250']
[YELLOW] BOM row 23 'Outlet header 219.1 OD x 8 thk' repeats item 2 (already covered by BOM row 7)
[BLUE] Suggestion (Cladding): the drawing notes imply it, but no matching line was found in the BOM

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.